

ARTICLES

Wavyleaf Basketgrass

This article describes wavyleaf basketgrass (*Opismenus undulatifolius*), an invasive grass species, and provides identification and management.

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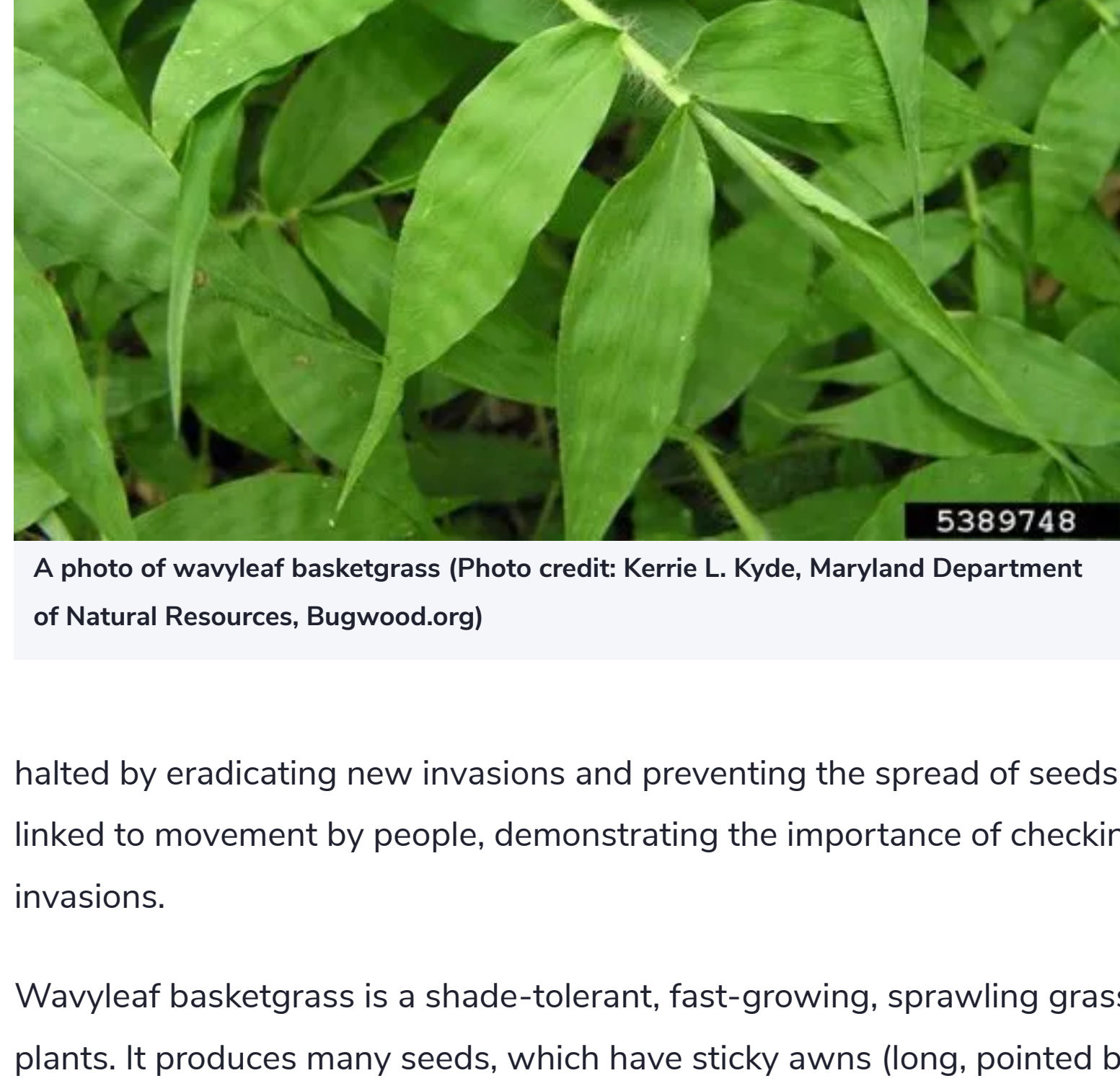
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A photo of wavyleaf basketgrass (Photo credit: Kerrie L. Kyde, Maryland Department of Natural Resources, Bugwood.org)

halted by eradicating new invasions and preventing the spread of seeds. In Pennsylvania, its spread has been linked to movement by people, demonstrating the importance of checking for seeds to help prevent new invasions.

Wavyleaf basketgrass is a shade-tolerant, fast-growing, sprawling grass that suppresses the growth of native plants. It produces many seeds, which have sticky awns (long, pointed bristles) that easily attach to boots, clothing, or the fur of anything that passes through an infested area. Seeds are easily distributed to new areas along trails or other high-traffic areas where they form dense new patches through creeping, above-ground roots (called stolons). To prevent its spread, it is important for anyone who encounters wavyleaf basketgrass to check for seeds on their clothing or pets. Remove any seeds and seal them in a plastic bag for disposal in a landfill.

Description

Size

Stems are short, low-creeping individuals that tend to be 8–11 inches tall, but may reach 18 inches tall, and grow in dense carpets.

Flowers

The feathery white flower spikes appear in late summer or fall (late August–November).

Seeds

Seeds grow in 3–5 pairs and have long awns, or stiff bristles, that secrete a sticky substance, allowing the seeds to easily stick to passing objects, people, and animals, even to smooth surfaces such as rubber boots.

Leaves

Leaf blades are deep green, lanceolate to ovate-shaped, and flat. Each leaf blade is between 0.5 and 1.0 inches wide and 1.5–4 inches long. The leaves have a distinct 'rippled' appearance, which earns the name 'wavyleaf.' Leaf blades are pubescent, with hair on the stems and leaf sheaths.

Stems

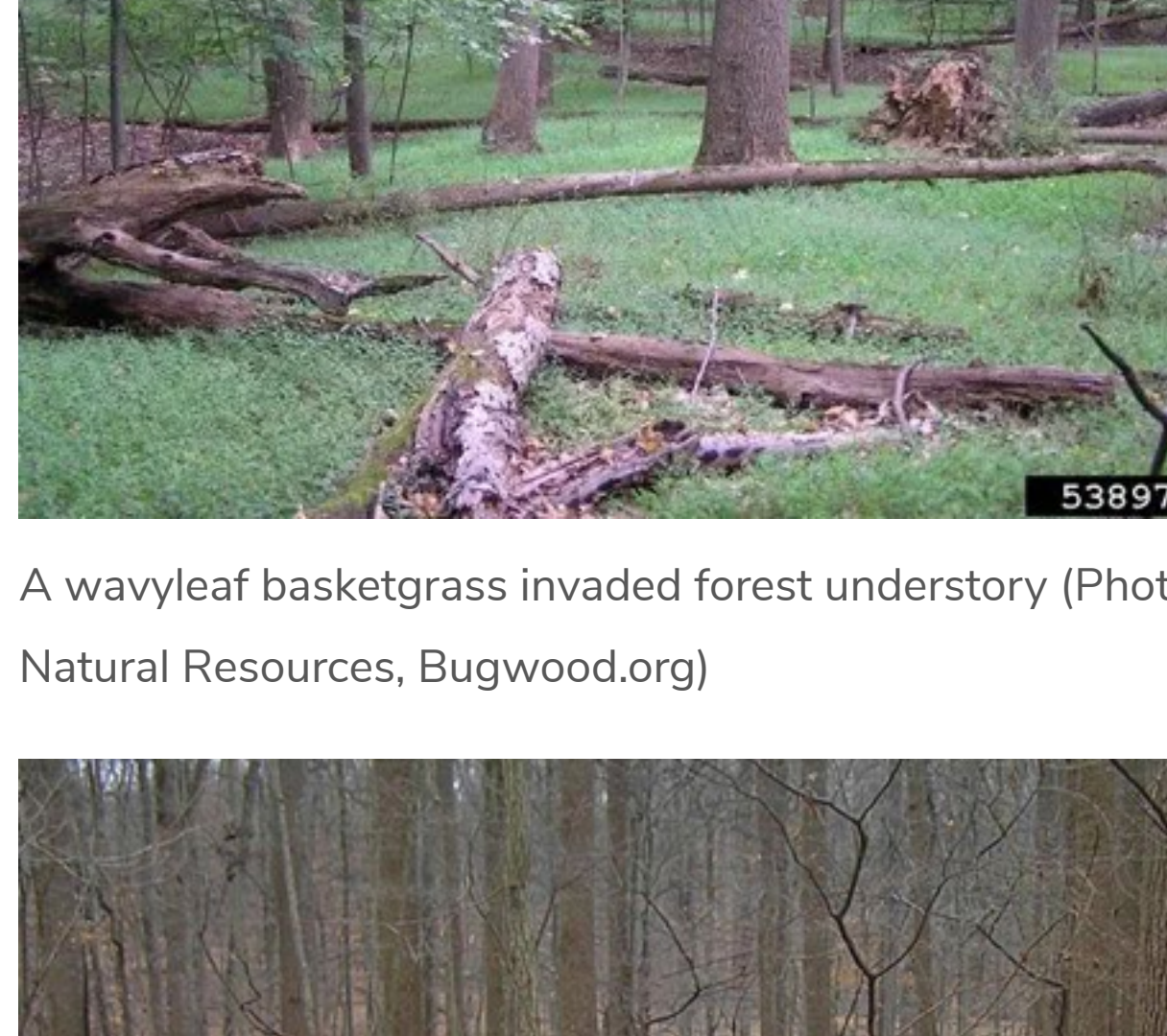
Stems are thin and pubescent with white hairs. They tend to be low-lying and root at nodes (where the leaves emerge).



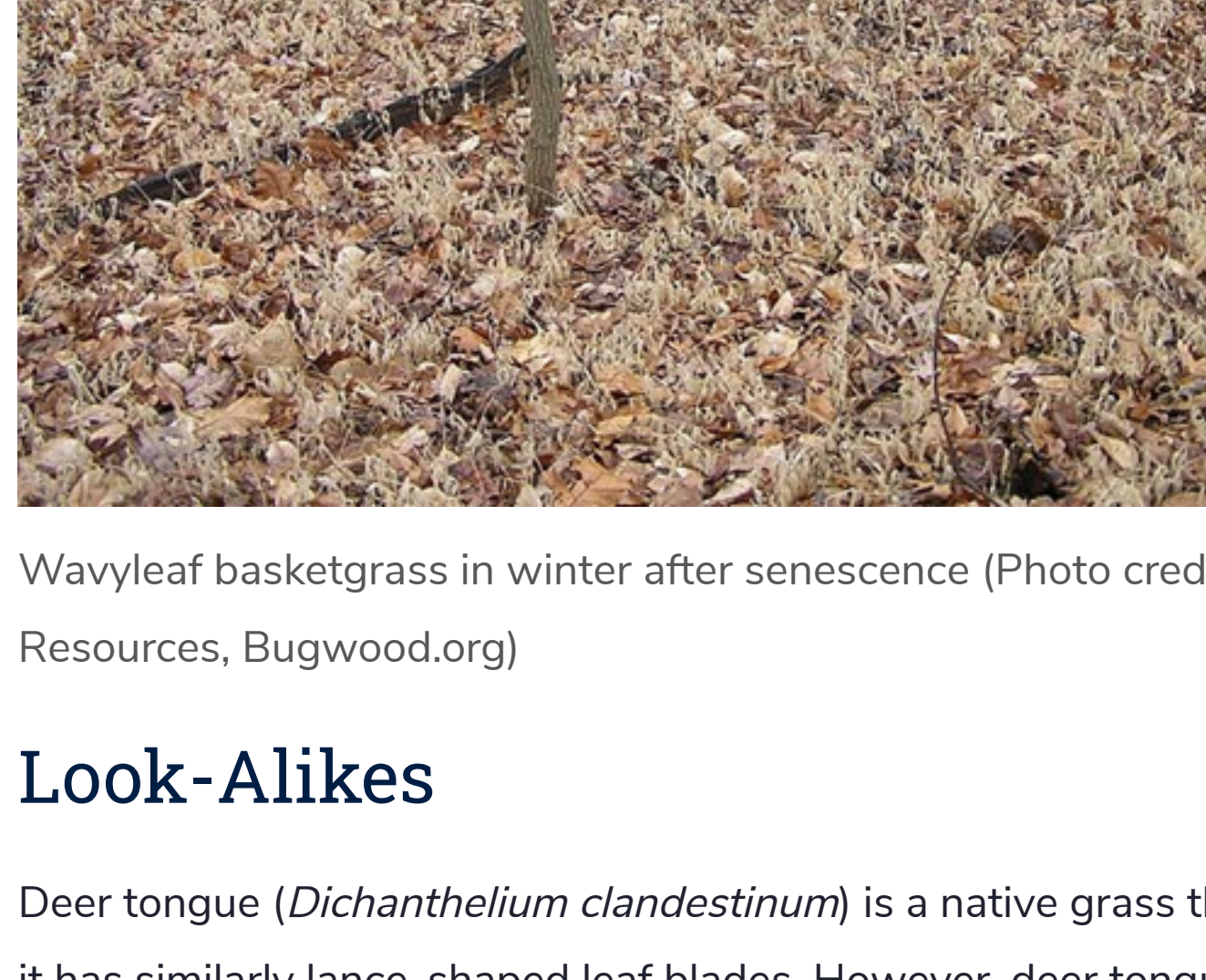
Wavyleaf basketgrass (Photo credit: Garrett Waugaman, M-NCPPC Weed Warriors, Bugwood.org)



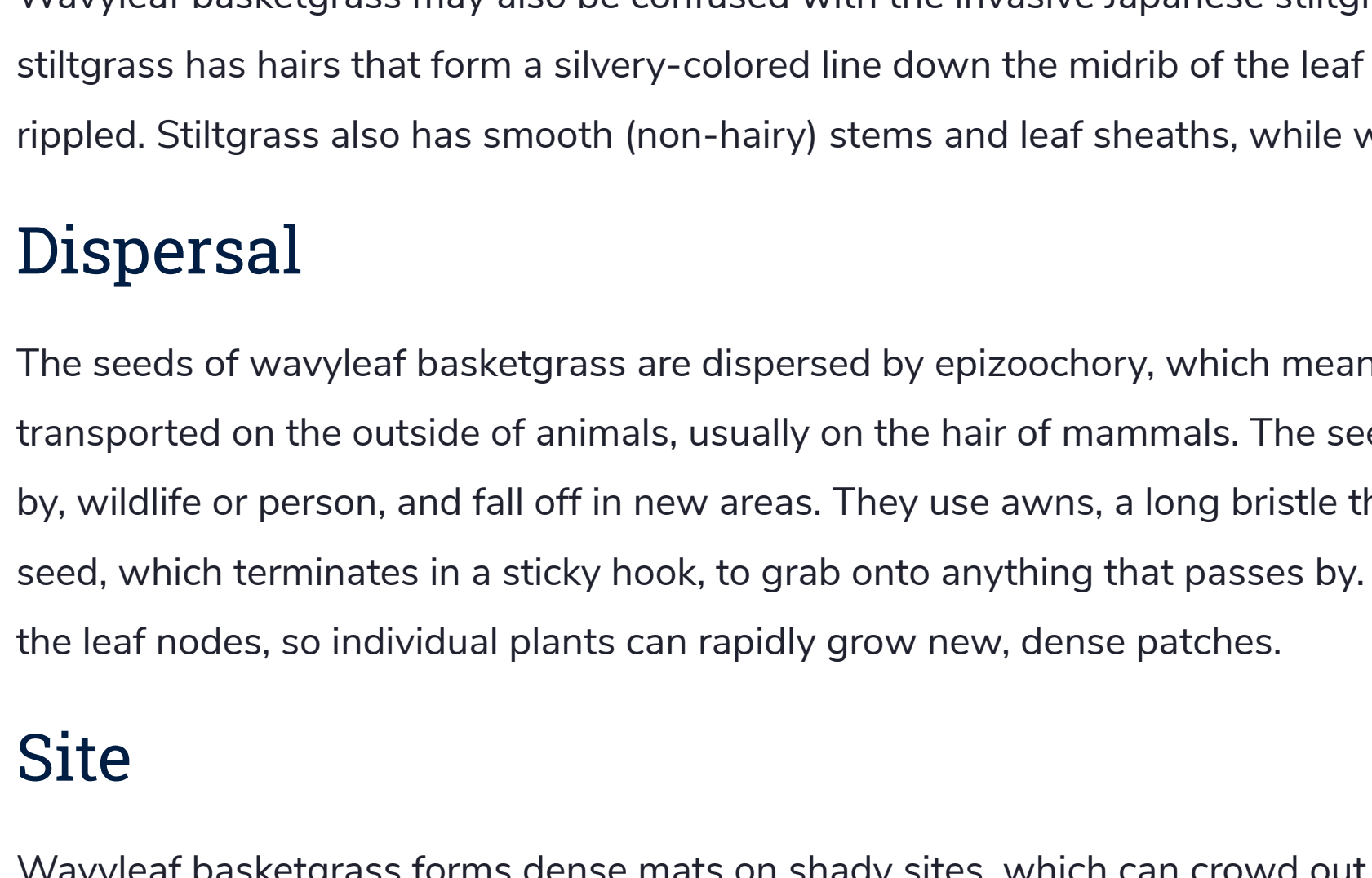
The flowers and seeds of wavyleaf basketgrass (Photo credit: Kerrie L. Kyde, Maryland Department of Natural Resources, Bugwood.org)



Leaves of wavyleaf basketgrass (Photo credit: Kerrie L. Kyde, Maryland Department of Natural Resources, Bugwood.org).



A wavyleaf basketgrass invaded forest understory (Photo credit: Kerrie L. Kyde, Maryland Department of Natural Resources, Bugwood.org)



Wavyleaf basketgrass in winter after senescence (Photo credit: Kerrie L. Kyde, Maryland Department of Natural Resources, Bugwood.org)

Look-Alikes

Deer tongue (*Dichanthelium clandestinum*) is a native grass that can be confused with wavyleaf basketgrass, as it has similarly lance-shaped leaf blades. However, deer tongue stems are semi-rigid and hairless; the leaves clasp the stem and are not rippled. Single plants tend to be larger than basketgrass, 3 1/2 to 4 1/2 feet tall. Deer tongue is also only semi-shade tolerant, so it is not likely to be found in deep shade where wavyleaf basketgrass thrives.

Wavyleaf basketgrass may also be confused with the invasive Japanese stiltgrass (*Microstegium vimineum*), but stiltgrass has hairs that form a silvery-colored line down the midrib of the leaf blade and the leaves are not rippled. Stiltgrass also has smooth (non-hairy) stems and leaf sheaths, while wavyleaf basketgrass is hairy.

Dispersal

The seeds of wavyleaf basketgrass are dispersed by epizoochory, which means the seed is dispersed by being transported on the outside of animals, usually on the hair of mammals. The seeds stick to anything that passes by, wildlife or person, and fall off in new areas. They use awns, a long bristle that extends from the end of the seed, which terminates in a sticky hook, to grab onto anything that passes by. Wavyleaf basketgrass roots at the leaf nodes, so individual plants can rapidly grow new, dense patches.

Site

Wavyleaf basketgrass forms dense mats on shady sites, which can crowd out numerous native plants (Figure 4). While it prefers shady areas, it may also be found in semi-shaded areas like trail edges.

Control

Due to wavyleaf basketgrass being a relatively new invader, there is limited experience in successful control, and research is ongoing. As a perennial, basketgrass cannot be controlled by mowing. This has been confirmed by its persistence in mowed disc golf courses. It is likely that mowing would spread invasions farther due to the likelihood of spreading seed or material on equipment rather than acting as a control. Hand-pulling small patches in the spring and summer can be used as control, but all plant roots need to be removed. When pulling, it is very important that all stems and roots are placed in sealable bags and disposed of in the trash or burned to ensure that control efforts do not inadvertently become spreading events.

At this time, two post-emergent herbicides—glyphosate and the grass-specific herbicide clethodim—are the only demonstrated effective controls for wavyleaf basketgrass. Post-emergent chemical control works by applying the herbicide directly to the foliage of actively growing plants. For treatments to be the most effective, herbicides should be applied in the spring or summer prior to flowering.

Clethodim is a grass-specific herbicide that will not harm sedges and broadleaf plants, while glyphosate is a broad-spectrum herbicide that will control all types of plants that come in contact with it. Both glyphosate and clethodim have been shown to be effective at controlling the majority of wavyleaf basketgrass infestations following one application. It often takes three years or more of application to eradicate entire patches of wavyleaf basketgrass. For improved control, it may be necessary to apply clethodim twice in one growing season to fully control an infestation. Applications of clethodim should take place at least three weeks apart and be completed before August, when flowering occurs.

Neither of these herbicides will control seeds stored in the seedbank, so patches must be monitored for several years following treatment. There is limited data on how long seeds can survive, but it appears that they can survive for at least a year, so patches should be monitored for at least three years. Herbicides may not be as effective if applied during periods of environmental stress, such as drought, or if it rains within an hour following application.

Early Detection

Since wavyleaf basketgrass is a relatively "new" invader and not yet widespread in the United States, there is hope for slowing or limiting its spread. Be on the lookout for this plant when walking in forests, parks, or other natural areas. It is critical to report any individuals that are found by taking good pictures and sending them to the local natural resource agency (the DCNR in Pennsylvania) or report it online through [iMap Invasives](#), [EDDMaps](#), or [iNaturalist](#). If infestations are detected early, control attempts are much more likely to be successful and less costly. This concept is called early detection and rapid response (EDRR), and it is key to preventing the spread of invasive species like wavyleaf basketgrass.

Management Calendar

The objective of wavyleaf basketgrass management is to eradicate new invasions and prevent seed production by controlling it early in the growing season. To ensure full eradication, the first treatment of clethodim or glyphosate would take place in June, with a second treatment in July prior to flowering and seed ripening in August and September.

	Jan.	Feb.	Mar.	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Germination												
Growth Initiation												
Hand Removal												
Postemergence Herbicide												
Flowering and Seed Ripening												

Herbicide Treatment and Timing

A colorant should be added to all herbicide mixtures to improve tracking, help avoid skips, and prevent duplicate treatments. Trade names are used here to provide specific information. Penn State Extension does not endorse or guarantee any product or recommend one product over another that might be similar. When products with the recommended active ingredients are available on Pennsylvania state contracts, those trade names are provided below. Other formulations with identical efficacy and label for forestry applications may be available.

When using herbicides, be sure to read and follow all the directions and safety precautions on the product label before mixing and using it. Be sure to handle products carefully and store them in the original labeled container that is out of the reach of children, pets, and livestock. When finished using, be sure to dispose of empty containers right away and in a safe manner. When working near water, be sure to use products labeled for aquatic use to avoid injuring aquatic organisms.

Treatment	Timing	Herbicide	Product Rates	Comments
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Postemergence Foliar	June through early-August (before basketgrass ripens seed)	Glyphosate Ranger Pro or Glyphosate Accord XRT II or Rodeo (aquatic-labeled)	4 quarts/acre or 3 quarts/acre	Currently, only post-emergence treatments are known to work as a control. Glyphosate is a nonselective herbicide with no soil activity. The products listed have different active ingredient concentrations and thus require different application rates. No additional surfactant is needed with Ranger Pro or Accord XRT II. If using a different glyphosate product, be sure to check the product label to see if a surfactant (e.g., CWC 90) is needed.
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Postemergence Foliar	June through early-August (before basketgrass ripens seed)	clethodim Envoy Plus*	16-24 ounces/acre	Clethodim is a grass-specific herbicide, meaning it will only control annual and perennial grasses; it does not control sedges or broadleaf weeds. Take care to avoid spraying native grasses. If using Envoy Plus, be sure to use a nonionic surfactant (e.g., CWC 90). Other clethodim products may also require a surfactant; be sure to check the product label.
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*Product contains a non-crop-site label. Be sure to check the product label to ensure the site to which you are applying is listed.

*There are no aquatic-labeled clethodim products

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